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Increased levels of harvest may favour sugar maple regeneration over American beech in northern hardwoods

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ABSTRACT

The increasing dominance of American beech regeneration (*Fagus grandifolia* Ehrh.) to the expense of sugar maple (*Acer saccharum* Marsh.) challenges the long-term economic viability of forest management in northern hardwoods. Based on earlier studies showing that more light in the understory favours sugar maple over beech, the irregular shelterwood system has shifted from experimental to operational over the last decade. In this paper, we evaluate the success of this shift toward irregular shelterwood for promoting sugar maple over beech by measuring the frequency, abundance of sugar maple and beech seedlings and saplings, and growth of seedlings three to six years after logging in northern hardwood forests of western Quebec, Canada. Results showed a dominance of beech regeneration regardless of the harvest intensity, particularly among tall seedlings and saplings. However, we found that the transition probability indices (projected relative combined abundance and growth) of sugar maple could be favoured, albeit to a limited extent, by an increased basal area removal, particularly where the initial occurrence of sugar maple in the advance seedling regeneration is lower than 60% and the initial beech basal area is low in the overstory (i.e. $< 6\ m^2\ ha^{-1}$). Our results highlight the importance of refining our operational management strategy to effectively limit the increasing abundance of regeneration of American beech over sugar maple in northern hardwood forests.

1. Introduction

The increase in abundance of American beech (beech, *Fagus grandifolia* Ehrh.) over sugar maple (*Acer saccharum* Marsh.) in the regeneration in northern hardwood forests of eastern North America in the last decades is challenging the management goal of production of high quality sugar maple timber. American beech is a less desirable species for timber production, which is also severely affected by the exotic beech bark disease, an infection by two fungi (*Neonectria faginata* and *N. ditissima*) that can be exacerbated by the presence of an exotic insect (beech scale, *Cryptococcus fagisuga* Lindinger) (Houston, 1994). The disease causes beech wood degradation and tree death without reducing the overall abundance of the species (Cale et al., 2017; Garnas et al., 2011; Houston, 1994). On the contrary, either as a reaction to the disease or to the consequent increased openings in the canopy, dense thickets of beech suckers and seedlings often develop in affected stands (Giencke et al., 2014; Houston, 2001; Roy and Nolet, 2018). Due to both its high shade-tolerance and ability to re-sprout from roots as suckers following disturbances, beech tends to be highly competitive in forests affected by small scale disturbances (Beaudet et al., 1999; Roy and Nolet, 2018). As a result, the factors negatively affecting the vigor and quality of beech also promote its regeneration, at the expense of both local biodiversity (Cale et al., 2013; Giencke et al., 2014) and a high yield of quality timber (Pothier et al., 2013).

Sugar maple is one of the most valuable species in northern hardwood forests and is highly sought after for so-called 'appearance' wood products such as furniture, cabinets and flooring as well as for maple syrup production. Sugar maple is the dominant species of a large bioclimatic zone in southern Quebec where it accounts for on average 15% to 25% of the growing stock volume (Boulet and Huot, 2013). Sugar

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maple prefers rich soil with moderate to good drainage, but it is highly adaptable (Boulet and Huot, 2013). Over the last few decades in Quebec, the species has experienced increased mortality, defoliation of the crown, and dieback resulting in decreased timber quality (Bal et al., 2014; Côté and Ouimet, 1996).

Although various environmental stressors have been identified as probable causes for this sugar maple decline (Bal et al., 2014; Bauce and Allen, 1991; Nolet and Kneeshaw, 2018), the increase of beech abundance in regeneration has also been considered as the predominant cause of sugar maple regeneration failure (Direction de la recherche forestière, 2017). The increasing abundance of beech can be partially explained by the fact that the species is often left behind by forest companies (Boulet and Huot, 2013; Majcen et al., 2003). In addition, damaged roots of residual beech trees resulting from logging operations may develop root sprouts that contribute to the regeneration success of the species (Jones and Raynal, 1988). Decreased light availability due to increasing beech sapling abundance limits the development of the sugar maple seedling bank, which is its main regenerative strategy (Beaudet et al., 1999; Hane, 2003; Marks and Gardescu, 1998).

In the absence of severe disturbances, the limited availability of light in the understory may alter the dynamics of sugar maple and beech regeneration. Both sugar maple and beech can survive for several decades in the understory (Canham, 1990; Poulson and Platt, 1996). Sugar maple and beech have a similar ecological niche; however, under a very dense canopy, beech performs better than sugar maple and has greater lateral growth (Beaudet et al., 1999; Burns et al., 1990; Canham, 1988; Poulson and Platt, 1996). The application of single-tree selection cuts in northern hardwoods results in small gaps and limited light availability in the understory, which is more favorable to beech than sugar maple regeneration (Bannon et al., 2015; Collin et al., 2017; Nolet et al., 2008; Poulson and Platt, 1996).

Accordingly, several studies have recommended that sugar maple regeneration could be favoured over beech in larger gaps or following a greater removal of the overstory canopy. Nolet et al (2008) indicated that a partial cut retaining more than 25% of the forest cover was not sufficient to favour sugar maple at the expense of beech although Dracup and MacLean (2018) suggested that removing ~ 45 % of basal area is sufficient to limit the conditions that are favourable for beech regeneration. Other studies have also suggested that the repetitive occurrence of large gaps may favour sugar maple over beech by providing higher light availability in the understory (Canham, 1988; Houston, 2001; Poulson and Platt, 1996). However, most of these results were obtained from carefully controlled experiments which may not be representative of the conditions created by silvicultural interventions conducted at the industrial scale (Puettmann et al., 2009).

Partial harvesting outcomes from controlled experiments may not be easily extended to commercial operations that are characterized by greater spatial heterogeneity of tree removal, and thus of light availability in the understory (Guay-Picard et al., 2015; Guillemette et al., 2013; Moreau et al., 2020). Further, the vast territory of Quebec's public forests creates a situation where there is no simple silvicultural solution to favour the sustainable establishment and recruitment of sugar maple. The currently recommended approach of mechanical removal of beech and fencing to protect against browsing (Bohn and Nyland, 2003; Bose et al., 2018; Nyland et al., 2006) is costly at the regional scale and may not be necessary in all situations. Glyphosate application, which can effectively control beech regeneration (Bose et al., 2018) is banned in Quebec's forests. Cost-effective, efficient solutions are needed to limit the establishment and recruitment of vigorous beech at the expense of sugar maple and maintain the economic viability of these commercially managed forests (Direction de la recherche forestière, 2017; Nelson and Wagner, 2014).

Because large openings may create more favourable regeneration conditions for sugar maple than beech, practitioners have recently switched from selection cutting to several variants of the irregular shelterwood system as their preferred management approach in northern hardwood forests (Ministère des Forêts, de la Faune et des Parcs, 2020, 2009). Variants such as continuous-cover or extended irregular shelterwood systems are more flexible regeneration systems that typically apply greater harvest intensities and more irregular cutting cycles than selection cutting (Larouche et al., 2013), which in turn may facilitate the establishment of species with different light requirements (Raymond et al., 2009). Whereas the selection system is characterized by cuts ranging from 20 to 30% in harvest intensity, irregular shelterwood systems generally rely on higher harvest intensities typically ranging from 30 to 50% ((Bédard et al., 2016; Larouche et al., 2013)). Despite being implemented in practice, no long-term studies have yet confirmed that this shift towards irregular shelterwood systems is efficient in favouring sugar maple regeneration over beech.

To rigorously test the hypothesis that the conditions created by a higher harvest intensity in partial cuts can help favour sugar maple regeneration over beech in commercially harvested forests, we resampled 96 plots three to six years after the application of various partial cutting treatments in the northern hardwood forest of Quebec, Canada. Our sampling plots were distributed among 12 harvesting sites where a range of treatments from single-stem selection to irregular shelterwood had been applied. In addition to assessing the current composition of the forest regeneration, we measured the growth of seedlings and projected growth over time using transition probability indices (Guay-Picard et al., 2015; Hill et al., 2005; Kneeshaw and Bergeron, 1998). These indices portray the short-term effect of harvesting by weighting a species abundance with its growth rate.

2. Material and methods

2.1. Study area

The study area is located in the northern hardwood forest of southwestern Quebec, Canada between longitudes 74° 30′ 0′'W and 76° 30′ 0′'W and latitudes 45° 47′ 60′'N and 46° 36′ 0′'N. Elevation in the region ranges from 270 to 440 m. The area is within the sugar maple – yellow birch (*Betula alleghaniensis* Britton) bioclimatic domain of Quebec's northern hardwood forest type. Mean annual temperatures in this domain vary between 2.5 and 4.0 °C (Saucier et al., 2009). Mean annual precipitation averages around 1000 mm (Saucier et al., 2009). Although it is dominated by sugar maple, the main forest canopy in the area is also composed of beech, yellow birch, red oak (*Quercus rubra*), balsam fir (*Abies balsamea* [L.] Miller), eastern hemlock (*Tsuga canadensis* [L.] Carrière), American hop-hornbeam (*Ostrya virginiana* [Mill.] K. Koch), and American basswood (*Tilia americana* L.). Approximately half of the study area experienced different intensities of forest tent caterpillar outbreaks in 2016, 2017 and 2018 (*Malacosoma disstria*). The insect feeds on codominant and dominant sugar maple tree leaves in northern hardwood forests, but can also eat other deciduous species except red maple (Boulet and Huot, 2013; Fortin and Mauffette, 2002; OIFQ 2009). A large part of the study area has undergone two or three commercial harvests over the last century. The study area is located across both the advance front and the killing front of the beech bark disease and damage was mostly prevalent in the southern part of the study area (Ministère des Forêts, de la Faune et des Parcs, 2019; Roy and Nolet, 2018; Shigo, 1972). Beech bark disease, or beech scale, was observed in half of our sites. The mean density of white-tailed deer (*Odocoileus virginianus*) was estimated as ranging from 2.26 to 3.05 per km^2 in 2017 for the whole study area (Ministère des Forêts, de la Faune et des Parcs, personal communication, January 11th, 2020) while mooses (*Alces alces*) are present, but less extensive (~0.24 per km^2 in 2002) in the study area (Ministère des Forêts, de la Faune et des Parcs, 2015).

2.2. Stand selection

Quebec's Ministry of Forests, Wildlife and Parks provided a database of available sampling plots established before harvesting. Twelve felling sites were first selected, each consisting of various partial cutting treatments conducted between 2013 and 2016 i.e. four sites with single-tree selection cuts, four sites with continuous-cover irregular shelterwood cuts and four sites with extended irregular shelterwood cuts. The extended irregular shelterwood method essentially consist of establishing and protecting a cohort of regeneration over a longer period than the standard shelterwood method. The continuous-cover irregular shelterwood method is more analogous to the selection method with the difference that both the harvest intensities and cutting intervals are expected to vary throughout a full rotation (See Raymond et al., 2009 of each variant). Throughout each felling site, plots had been established systematically every 150 to 200 m prior to harvest. We first extracted all plots from the database in which either single-tree selection, continuous-cover irregular shelterwood or extended irregular shelterwood cuts had been conducted between 2013 and 2016, with at least 25% of the basal area consisting of either sugar maple or beech before treatment, and with good or moderate drainage and with at least 50-cm-thick soils of glacial or fluvio-glacial origins, as mapped by the Ministry with photo-interpretation. These soil characteristics are the most common in the study area. We then randomly selected eight plots per harvesting site. Except when restricted by regulations or where terrain limited access, all plots had been harvested with fully mechanized operations. Harvesting operations were conducted either in the late summer, fall or winter. A total of 96 plots were sampled during the summer of 2019 (Fig. 2.1).

2.3. Data collection

2.3.1. Pre-treatment sampling

The sampling plots database provided by the Ministry contained the pre-treatment data. Variable-radius plots were established with a basal area prism of a factor of 2. All trees with a DBH>9 cm were tallied. Pre-treatment regeneration sampling considered seedlings with minimum heights of either 15 cm or 40 cm, depending on the sites. In the Outaouais region, a cluster of five, fixed-radius subplots was established, one in the center of the main plot and four others in the cardinal directions, 10 m from the center. Following provincial standards, the subplot radius depended on the species sampled (between 1.13 m and 2.82 m). The regeneration sampling design in the Laurentides region also consisted of five fixed-radius subplots, using the same species-specific radii. However, in this region the first plot was established 6 m from the center in the direction of the next plot and the four others were successively established at a 6-meter interval in the same direction. In both regions, the presence of seedlings was recorded for each species to indicate occurrence, but not abundance. The occurrence refers to the percentage of subplots in which a seedling of a given species and class height was found.

2.4. Post-treatment sampling

Our post-cut sampling was conducted from late May until August 2019. First, from the center plot we updated the pre-harvest data in the main plot for all trees with a DBH larger than 9 cm, with the identification of live and dead trees as well as stumps to confirm removal and obtain the post-treatment basal area. To assess the regeneration, we reproduced the same regeneration sampling design as in the Outaouais region and maintained the same method of distributing subplots in a cross pattern instead of in a line as in the Laurentides (Fig. 2.2). In all cases, it was impossible to make sure the regeneration subplots were located at the exact same position as in the pre-harvest assessment. A member of the fieldwork team estimated canopy openness using a forest densiometer calculating an average of five measurements for each

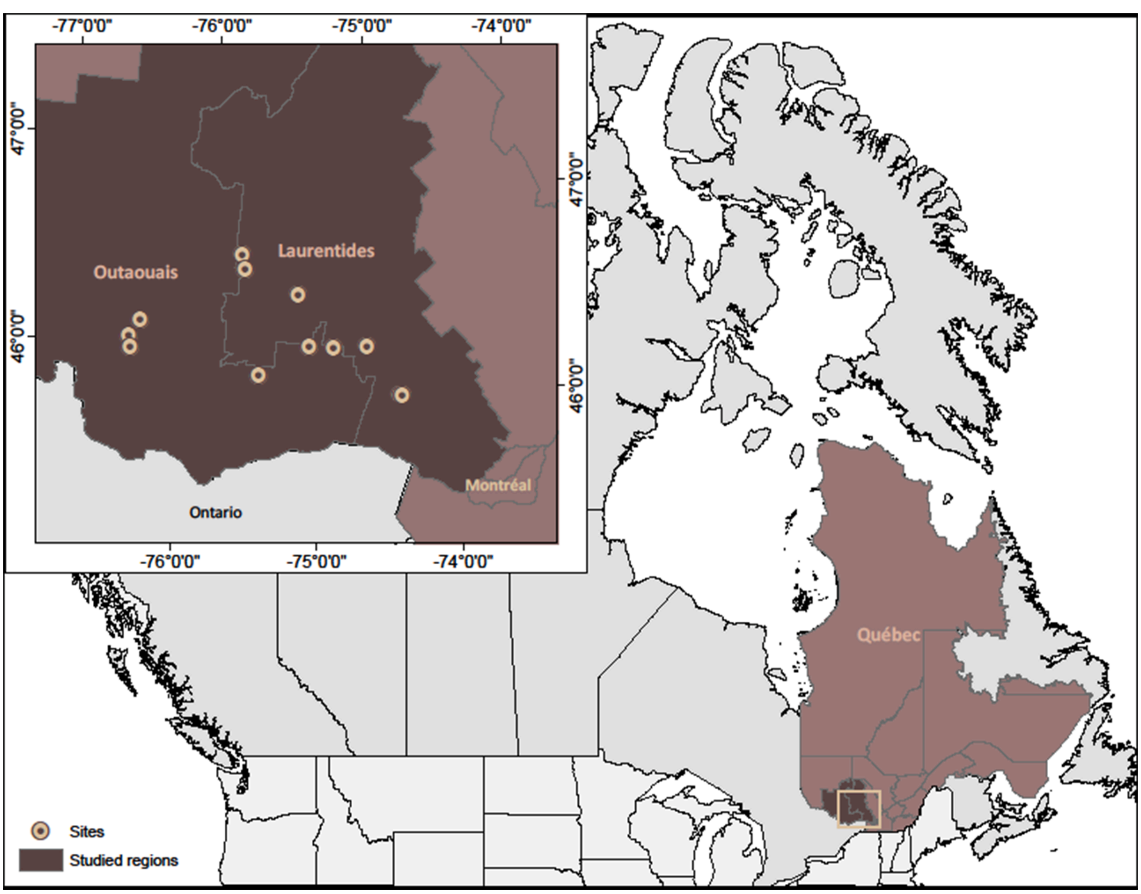

Fig. 2.1. Study area and location of the 12 harvesting sites.

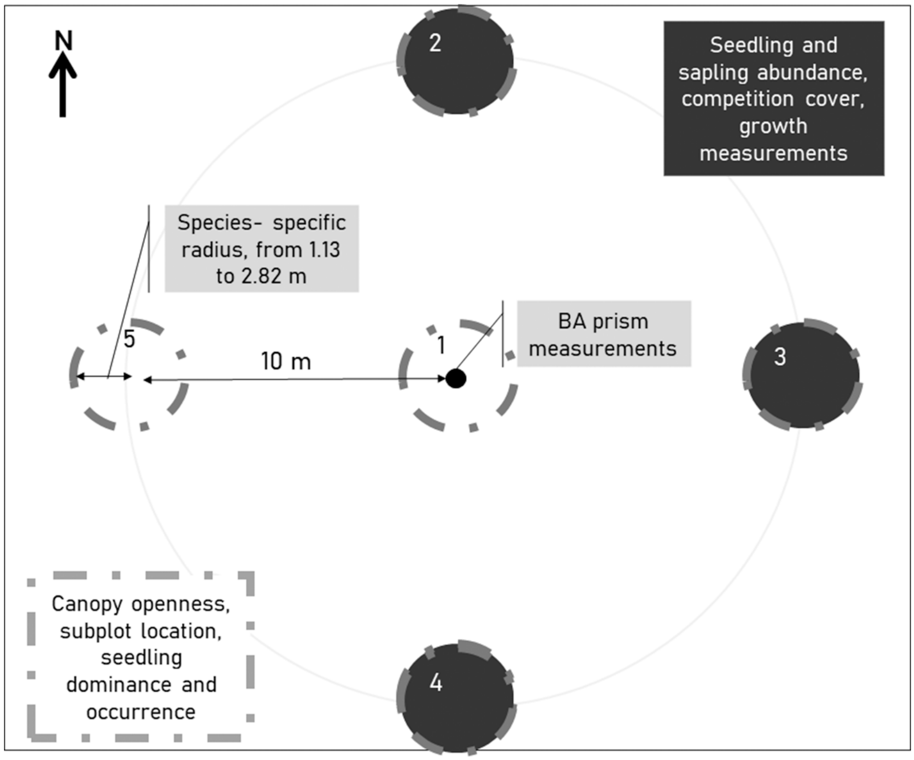

Fig. 2.2. Variable-radius plot design. In all subplots (grey dotted outline), we marked the subplot location and measured canopy openness, seedling dominance and occurrence. In subplots 2, 3 and 4 (black filling), we measured seedling and sapling abundance, competition cover and seedling growth. We measured basal area (BA) from the center of the main plot with a prism of factor 2.

subplot, at a height of 1.3 m. Therefore, unlike basal area removal, canopy openness measures were different for each subplot and were taken 3 to 6 years after cut. Also, we identified whether the subplot was mostly located in the matrix (i.e. uncut or partially cut area of the harvest site where the harvester can circulate) or in a skid-trail (i.e. where most of the trees are cut to allow log extraction). The seedlings were divided into two height classes i.e. small seedlings from 15 to 40 cm in height and tall seedlings from 40 cm in height to a DBH of 1 cm. We also tallied saplings with their DBH (DBH > 1 to 10 cm). The presence of each species for each height class was recorded in each subplot. The species that would most likely dominate the seedling class and the subplot class was noted separately for each subplot. We defined dominance as the tallest vigorous stem in its respective class. If equally dominant, up to three species were identified in this assessment. Finally, we identified the competition cover (0, >0–25%, 25–75 %, 75–100%) for each regeneration height class. We considered as competition vegetation any non-merchantable species that had reached at least 15 cm in height. The competition cover classes were cumulative i.e. the cover of saplings considered as competition (e.g. striped maple (*Acer pensylvanicum*)) would affect all equal and lesser height classes. Other examples of competition species were *Rubus idaeus*, *Sambucus pubens*, *Viburnum lantanoides* and *Dryopteris carthusiana*. In three of the five subplots (north, south and east), we counted non-browsed seedlings per height class and per species. Preferential browsing affects northeastern hardwood forest succession as white-tailed deer and moose prefer sugar maple over beech (Bose et al., 2018; Elenitsky et al., 2020; Matonis et al., 2011). We considered a seedling to be browsed if it had been damaged over the last 3 years, as indicated by the subsequent primary growth of the seedling. In subplots with extreme seedling densities, from one-fourth to three-fourths of the subplot was sampled for efficiency, and estimates were scaled up to that of the entire plot. We measured the annual height growth of seedlings for the previous three growing seasons, using scale scars. For coniferous species, except eastern hemlock and eastern white cedar (*Thuja occidentalis*), we used the annual whorls of branches. We randomly chose two samples per species per height class and per subplot to measure the annual height growth. Again, we did not consider browsed seedlings because we wanted three full years of growth. Growth of the current year was not considered since the assessments were made at different times throughout the summer.

2.5. *Data analysis*

2.5.1. *Annual height increment*

To avoid limiting our assessment to current conditions and to take into account the dynamic nature of regeneration establishment and growth, we used the concept of transition probabilities first proposed by Kneeshaw and Bergeron (1998) and later applied by Hill et al. (2005) and Guay-Picard et al. (2015). The first step of this method consists of predicting the height growth of seedlings. For this, we produced an annual height increment linear mixed-effects model for the five most abundant species in our study sites i.e. sugar maple, beech, yellow birch, American hop-hornbeam and red maple, as well as for all the other species pooled together. These groups were included as a six-level categorical variable in the model. Only the non-browsed seedlings were considered. The model was fitted using the *nlme* package (Pinheiro and Bates, 2011) in the R statistical programming environment (R Core Team, 2018). To limit the influence of inter annual variability, height growth assessments consisted of the mean of the last three years. For this, we only used the data from sites that were harvested in 2013 and 2014 (24 subplots from eight felling sites) to average three full years in post-treatment conditions only. Site, plot and subplot levels were considered as nested random effects. A set of seven candidate explanatory variables was defined *a priori*, namely species, seedling height, subplot location (matrix or skid trail-located), subplot canopy openness, subplot competition cover class, subplot sapling basal area and subplot seedling abundance of all species except that of the subject. We also included three possible interactions in the full model (species: seedling initial height, canopy openness: subplot location and canopy openness:

competition class). The initial height of the seedling was selected as a covariate for its known effect on seedling growth (Guay-Picard et al., 2015). The other variables were linked to understory resource availability, which can also influence seedling growth (Beaudet et al., 2007; Canham, 1988; Hannah, 1991). The best model was selected based on the results of likelihood-ratio tests for nested models (Pinheiro and Bates, 2011). For all analyses, residuals were analyzed to ensure that statistical assumptions were met. When outliers were identified, the results of the analysis were examined without them to ensure that any outliers were not a large source of bias. All outliers were maintained.

2.5.2. Transition probability indices

Instead of giving a static picture in time, transition probabilities portray regeneration using the current abundance and growth rate of each species (Guay-Picard et al., 2015; Hill et al., 2005; Kneeshaw and Bergeron, 1998). Transition probabilities are in fact indices of the projected relative abundance, in this case expressed at the subplot level. The method provides indices of forest succession, in this case in the regeneration strata, but does not take into account mortality and recruitment after data sampling. Using height growth predictions from the model described above, we used the relative time to reach 2 m to weight the seedling density in a given height class. For this, we predicted annual height increments for the mean initial height of each height class (27.5 cm for small seedlings and 120 cm for tall seedlings) of each species which was present in any given subplot. We then integrated the annual growth increment function to obtain the predicted time for the seedling to reach 2 m and normalized the data by calculating the ratio of time to reach 2 m for a given species in a given height class to that of the slowest growing species. The reference of 2 m was chosen as an indication of regeneration reaching the sapling stage. Transition probability indices were then calculated using:

$$P_t = \frac{\sum_{i=1}^{2} (RA_i/\Delta T_i)}{\sum_{t=1}^{n}\sum_{i=1}^{2} (RA_i/\Delta T_i)_n} \tag{1}$$

where P_t is the transition probability index for a given species t (t = 1,2, …,n), RA_i is the relative abundance of species t in height class i (i = 1 (small seedlings), 2 (tall seedlings)) and ΔT_i is the relative time necessary to reach 2 m for species t in height class i. The sum of transition probability indices within a subplot is 1. For a given species, transition probability indices can therefore vary between 0 and 1 with several occurrences of 0 and 1 values. Accordingly, transition probability indices were modelled using a zero-one inflated beta (BEINF) regression approach. We used the *gamlss* package (Rigby and Stasinopoulos, 2005; Stasinopoulos et al., 2017) in the R statistical programming environment (R Core Team, 2018) to run the analyses. As explained by Durocher et al. (2019), the BEINF() function defines the beta-inflated distribution with a four-parameter distribution where μ describes the vector of location parameter values > 0 and < 1, σ the vector of scale parameter values > 0 and < 1, ν the vector of parameter values associated with the probability at zero and τ the vector of parameter values associated with the probability at one.

As a data exploration step, we first fitted a full model with a set of candidate variables to understand which factors related to silviculture could better explain the transition probability indices obtained for sugar maple. Priority was given to candidate variables that were easily measurable in the field. We first selected the pre-treatment occurrence of sugar maple in the advance seedling regeneration (separated into three ordered factors: 0% (absent), 20–60% (low) and 80–100% (high)) as a covariate likely related to the transition probability values (Guay-Picard et al., 2015). Since they are manageable as part of commercial forest operations, we added variables linked to light availability, that is, the percentage of basal area removal at the plot level and the canopy openness measured by the densiometer at the subplot level. We then added the pre-treatment basal area per hectare in beech as a candidate variable since mature beech trees tend to produce root suckers and clump shoots when cut (Burns et al., 1990). We also added the calendar year of the first growing season after treatment as a candidate variable since it can be related to factors such as climatic conditions, forest tent caterpillar defoliation or seed abundance conditions. Finally, because the application of a silvicultural treatment can vary between locations and operators, and because the silvicultural prescription was likely related to the pre-harvest composition and state of the regeneration, we chose not to include the prescribed type of partial cut among the explanatory variables. Pearson's r coefficients between candidate variables were consistently below $|0.4|$.

To reflect the structure of the variance, we aimed to insert nested random effects of the site and the plot to the four components of the model (μ, σ, ν and τ), but the full model had to be simplified to allow fitting. First, only the random effects of site were fitted in the full model for components μ, ν and τ, and then the nested subplot effect was added at the end of the variable selection process.

We based our model selection method on Stasinopoulos et al. (2017) and used a backward elimination by AIC with the *gamlss* package. The backward elimination strategy was motivated both by its simplicity and the fact that all candidate variables could plausibly be expected to be related to the response. We first applied the backward selection to the full model for τ. Then, using the model obtained for τ, we applied the same backward selection process to the ν mode, and successively applied the same process for σ and μ. Statistical assumptions related to residuals for GAMLSS such as homogeneity of variance, normality skewness and kurtosis were checked. Using the final fitted model, we predicted transition probability indices from two simulated databases. For this, we created classes representative of the observed variability for each variable except basal area removal and canopy openness, which were correlated variables. In theses cases, we created classes for one while using the mean value of the other for each class.

3. Results

3.1. Pre- and post-harvest attributes

Prior to treatment, the surveyed plots were mainly composed of sugar maple. The mean initial basal areas of sugar maple and beech were 14.9 and 4.2 m^2 ha^{-1}, respectively. The initial basal area of beech in our plots varied from no beech stems in the plots to a maximum of 22 m^2 ha^{-1}. The applied partial cuts reduced the initial stand basal area from 24.8 to 14.7 m^2 ha^{-1} (~40% average reduction ranging from 0 to 80%). There was a high variability among plots and subplots in terms of canopy openness, proportion of basal area removed and final basal area not only among the types of partial cuts, but also between plots with each type of cut (Fig. 3.1).

3.2. Regeneration characterization

3.2.1. Occurrence, abundance and dominance

Prior to harvesting, the mean occurrence of species in the advance seedling regeneration of the surveyed subplots (as provided, combining small and tall seedlings) was 61% (SD = 39%) for sugar maple compared to 40% (SD = 41%) for beech. Three to six years after harvesting, sugar maple had maintained a higher mean occurrence than beech in the small seedling class with 77% (SD = 31%) compared to 56% (SD = 37%) for sugar maple and beech, respectively. However, in the tall seedling class the occurrence was similar between species, with 61% (SD = 39%) compared to 64% (SD = 37%) for sugar maple and beech, respectively. The most abundant species within the small seedling class was sugar maple with mean densities of approximately 12 000 stems ha^{-1} (Fig. 3.2). The abundance of sugar maple decreased sharply in the tall seedling class, so that beech became the most abundant species with densities averaging 3200 stems ha^{-1}. beech saplings were also the most abundant. beech was the most common dominant species in the two seedling classes combined as it dominated 232 of the 480 subplots

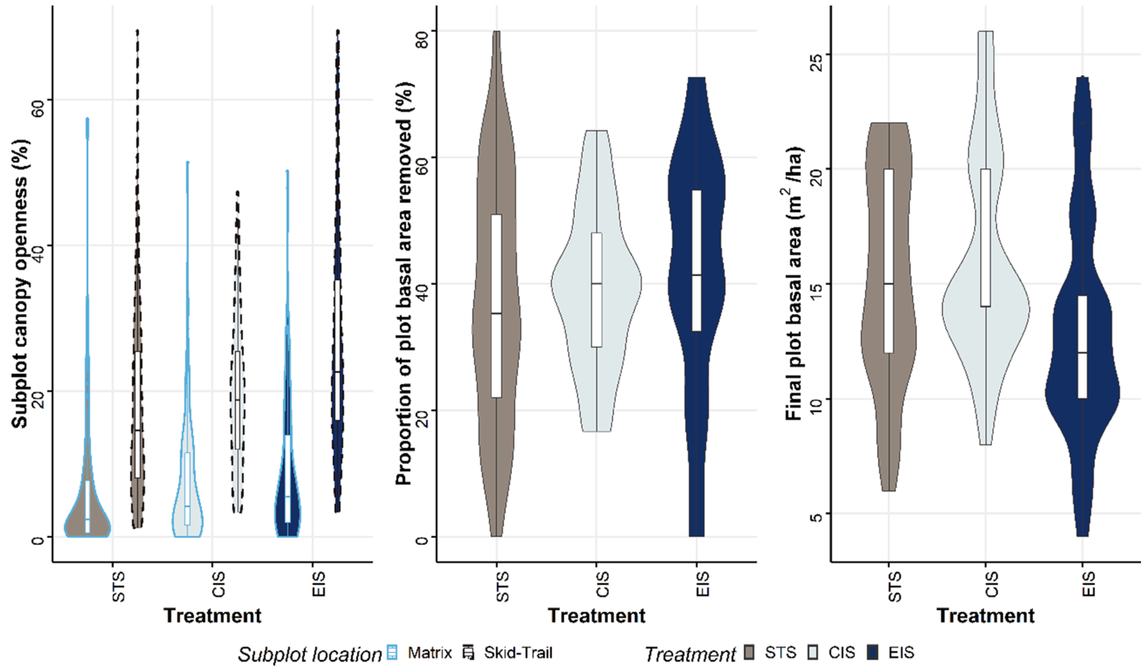

Fig. 3.1. Attributes related to understory resource availability per treatment. STS: Single-tree selection cut, CIS: continuous-cover irregular shelterwood cut, EIS: extended irregular shelterwood cut. For subplot canopy openness, difference between matrix (light blue solid line) and skid-trail (black dashed line) is represented. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

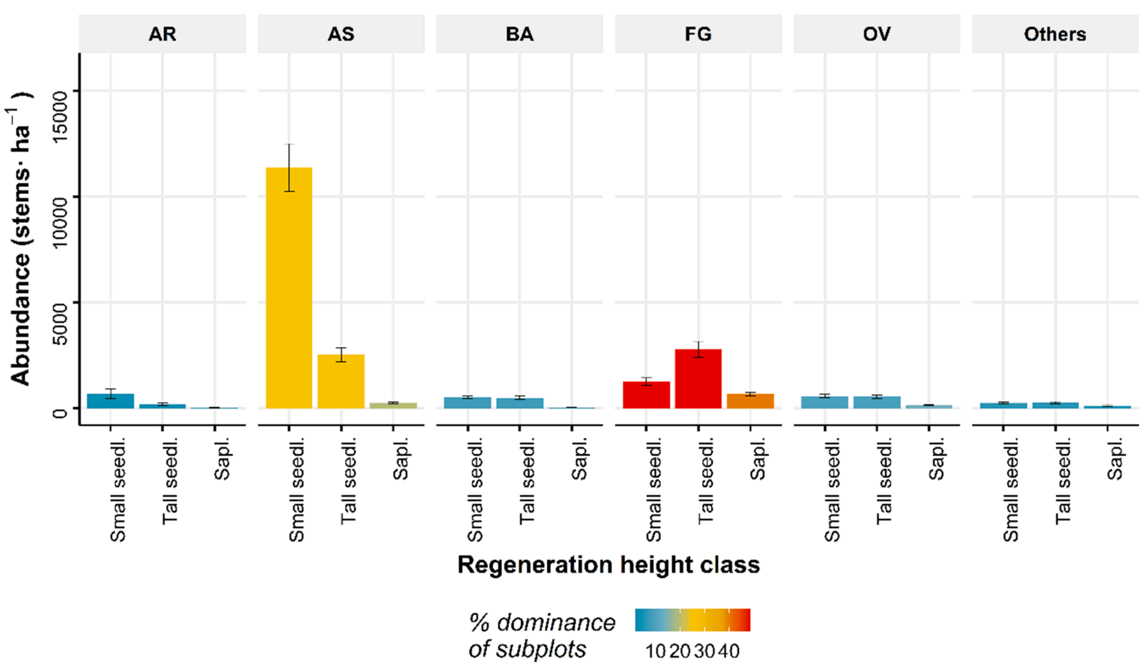

Fig. 3.2. Abundance (mean + SE) and dominance of regeneration. Dominance was observed for seedlings (two classes combined) and for saplings and is defined as the tallest vigorous seedling. AR. = *Acer rubrum*, AS. = *Acer saccharum*, BA = *Betula alleghaniensis*, FG. = *Fagus grandifolia*, OV = *Ostrya virginiana*. Seedl. = seedlings, Sapl. = saplings.

sampled (48%). beech saplings also dominated 40% of the subplots. By comparison, sugar maple seedlings and saplings dominated only 25% and 19% of the measured subplots, respectively.

3.2.2. *Growth and relative growth*

The final annual growth model included the subplot location (matrix or skid-trail), the canopy openness, the species, the seedling height and the interaction of the latter two as explanatory variables. Small sugar maple seedlings located in the forest matrix with 0% canopy openness were predicted to take the longest time to reach the 2 m threshold (over 36 years). We considered these seedlings as the reference to calculate the relative growth for seedlings in all other conditions. In comparison, the fastest growing seedlings were predicted to be tall yellow-birch seedlings located in a skid-trail with almost 70% of canopy openness that were predicted to reach the 2 m threshold in approximately 1.6 years (Fig. 3.3). Moreover, the fastest sugar maple seedlings were predicted to

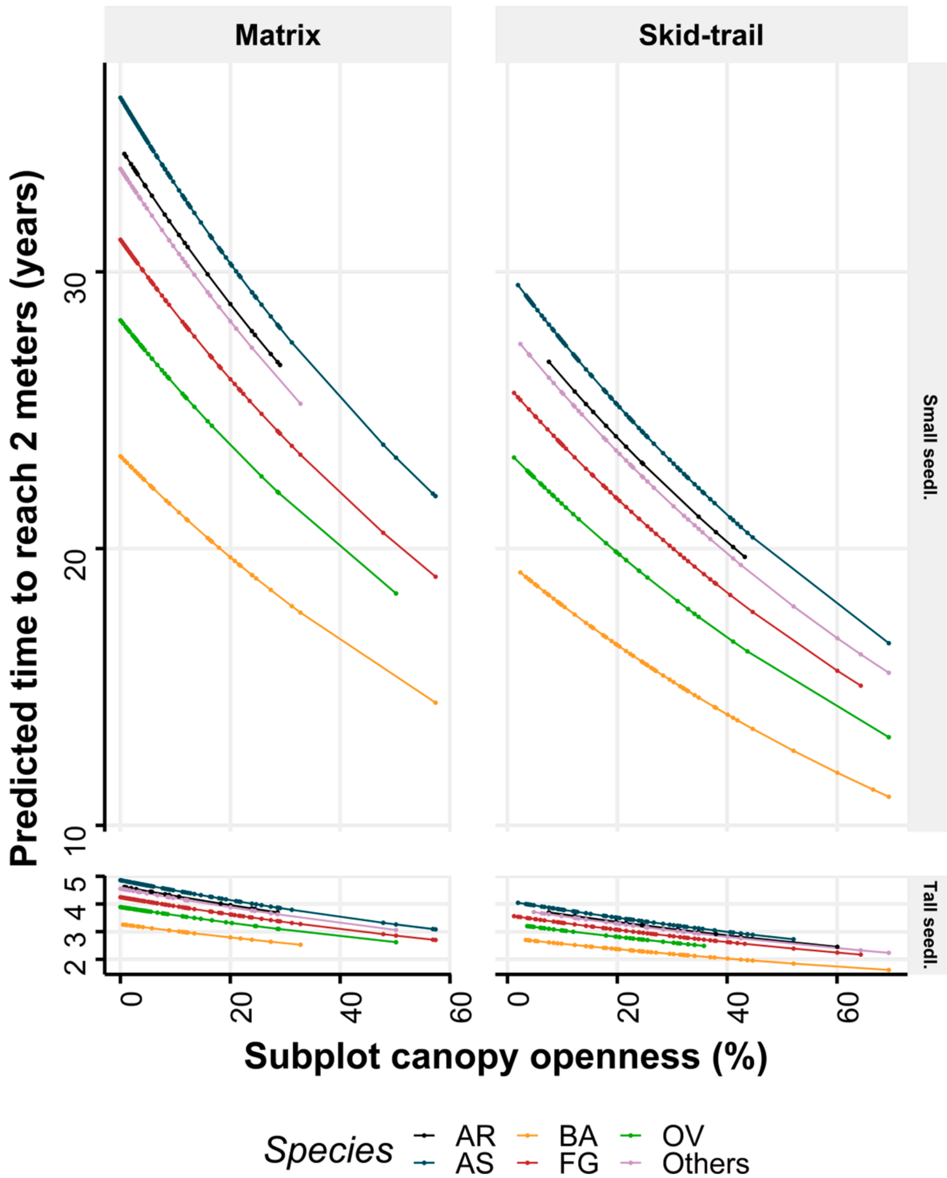

Fig. 3.3. Predicted time to reach two meters in height as a function of measured subplot canopy for five hardwood species and other species pooled together, for two height classes (small and tall seedlings) and for two subplot locations (skid-trail or matrix). Tall seedlings are all considered 120 cm high and small seedlings 27.5 cm high. AR. = *Acer rubrum*, AS. = *Acer saccharum*, BA = *Betula alleghaniensis*, FG. = *Fagus grandifolia*, OV = *Ostrya virginiana*.

be tall seedlings located in a skid-trail with a subplot canopy openness of just over 50%, which were expected to reach 2 m in 2.7 years. In comparison, the slowest beech seedlings were predicted to be small seedlings growing in the forest matrix with 0% canopy openness, where were expected to reach 2 m in 31 years. The fastest beech seedlings were predicted to be tall seedlings in a skid-trail with 64% canopy openness, for which the anticipated time to reach 2 m was 2.1 years.

3.2.3. Transition probability indices

A high level of variation in transition probability indices was observed among the sites, plots and subplots (Fig. 3.4). The highest transition probability values were generally associated with either beech or sugar maple. All other groups had much lower results with a median index at zero in all cases.

Five explanatory variables and two nested random effects contributed to the most parsimonious model for sugar maple transition probability indices (Table 3.1). Based on this model, transition probability indices for sugar maple tended to increase with the pre-treatment occurrence of the species in the advance seedling regeneration, while indices decreased with an increasing initial basal area in beech. An increase in the proportion of the initial basal area removed (intensity harvest) was associated with a lower probability of a 'zero transition probability' for sugar maple. Predictions combining the zero and non-zero components of the *gamlss* fitted model are shown in Figs. 3.5 and 3.6. Transition probability indices for sugar maple increased slightly with increasing plot basal area removal although the effect was reduced when beech initial basal area was high. An increase in the occurrence of sugar maple in the advance seedling regeneration (i.e., prior to treatment) also favoured sugar maple transition probabilities indices significantly. However, the impact of basal area removal appeared to decrease when the occurrence of sugar maple in the advance seedling regeneration was high. Overall, the difference in transition probability indices for sugar maple between a plot basal area removal of 80% and 25% (maximum and minimum subplot intensity harvest) reached a maximum of around 0.1 when the initial basal area in beech was zero and the occurrence of sugar maple in the advance seedling regeneration was 60% or less. The effect of subplot canopy openness on the transition probability indices of sugar maple was less pronounced than that of

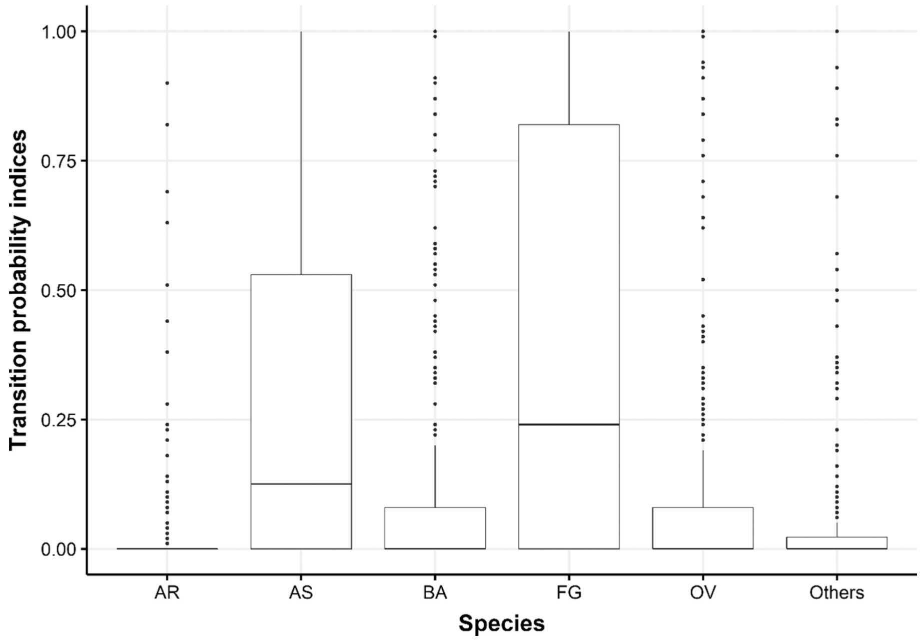

Fig. 3.4. Distribution of transition probability indices per species. Transition probability indices combine relative abundance and relative growth of every species in a subplot. The sum of transition probability indices within a subplot is 1. For a given species, the transition probability index can therefore vary between 0 and 1 with several occurrences of 0 and 1 values. AR. = *Acer rubrum*, AS. = *Acer saccharum*, BA = *Betula alleghaniensis*, FG. = *Fagus grandifolia*, OV = *Ostrya virginiana*.

Table 3.1
Details of the fitted gamlss model. Final model for sugar maple. μ describes the vector of location parameter values > 0 and < 1, σ the vector of scale parameter values > 0 and < 1, ν the vector of parameter values modelling the probability at zero and τ the vector of parameter values modelling the probability at one. BA : basal area , FGY: Year of the first growth season after harvest. S.E. is the standard error of the parameter estimates.

Dependent variable	DF	Component	Explanatory variables	Significant parameters	Parameter estimates	S. E.	P. value
Sugar maple transition probability indices	117	μ	Pre-treatment sugar maple occurrence + Initial beech BA + Site/Plot nested random effects	Intercept	−0.35	0.09	<0.001
				Pre-treatment sugar maple occurrence (linear) *	0.46	0.14	0.001
				Initial beech BA	−0.08	0.01	<0.001
		σ	FGY + Initial beech BA	Intercept	−0.33	0.13	0.01
		ν	Canopy openness + Proportion of total initial BA removed + Pre-treatment sugar maple occurrence + Site/Plot nested random effects	Intercept	−0.34	0.49	0.49
				Proportion of total initial BA removed	−0.02	0.01	0.0519
				Pre-treatment sugar maple occurrence (linear) *	−2.13	0.39	<0.001
		τ	Canopy openness + Proportion of total initial BA removed + FGY	Intercept	−2.98	1.05	0.005
				Initial beech BA	−0.47	0.23	0.045
				FGY (2015)**	0.86	1.18	0.46
				FGY (2017)**	2.55	1.13	0.025

* Pre-treatment sugar maple occurrence is a factor set as ordered (0% (absent), 20–60% (low) and 80–100% (high)). Thus, the results showed a significant linear trend of pre-treatmeant sugar maple occurrence on sugar maple transition probability indices.

** FGY (2015) and FGY (2017) represent two levels of a categorical variable. A non significant effect is shown for level FGY (2015) because only the effect of FGY (2017) was found to be statistically significant. Level FGY (2014) was used as the reference so its effect is included in the intercept.

basal area removal at the plot level. Moreover, the effect tended to become saturated at a subplot canopy openness of approximately 50%.

4. Discussion

4.1. *Sugar maple-beech regeneration dynamics*

Despite the dominance of sugar maple in the pre- and post-harvest overstory and its high abundance of small seedlings, sugar maple seedlings were dominated by beech overall after harvest over the full range of harvesting conditions investigated. This trend became progressively stronger as regeneration size increased. Beech seedlings dominated almost 50% of the sampled subplots and had the highest mean transition probability indices among all species. As observed by Elenitsky et al. (2020) after beech bark disease-motivated harvests in northern Michigan, sugar maple regeneration was less dominant than that found prior to harvest. Such a dynamic occurring in northern hardwood forests has been well documented both for old-growth and managed stands (Bose et al., 2017; Duchesne et al., 2005). Based on this study and also that observed by Nolet et al. (2008), increasing harvest intensity in partial cuts in a commercial context appears to be generally insufficient to favour the regeneration of sugar maple over beech. While the ecological niche of sugar maple and beech is very similar, the ability to produce root suckers likely leads to beech's competitive dominance (Beaudet et al., 1999).

However, greater basal area removal did have a significantly positive

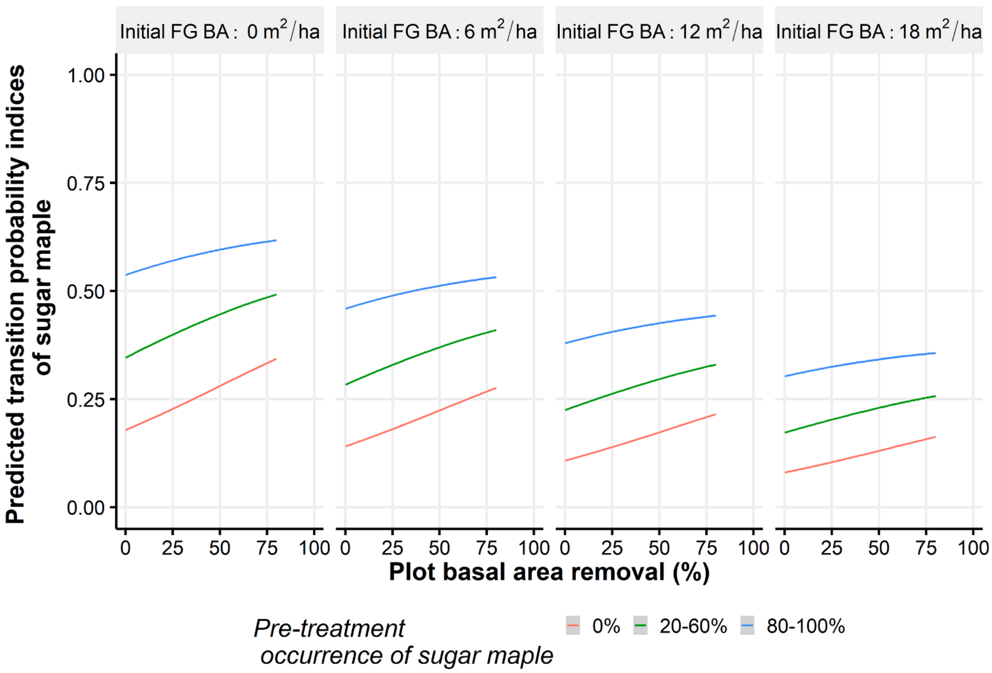

Fig. 3.5. Predictions of transition probability indices for sugar maple as a function of plot basal area removal per class of sugar maple pre-treatment occurrence in the advance seedling regeneration and intial beech basal area (FG BA). Because we predicted transition probabilities from a simulated database, no confidence intervals are presented. We divided the observed range of basal area removal (0 to 80%) into four sections and calculated the mean canopy openness for each. For a simulated basal area removal of 0 to 20%, the mean canopy openness was 8%. For a simulated basal area removal of 20 to 40%, 40 to 60% and 60 to 80%, the mean canopy openness was 10%, 15% and 27%, respectively. For every jump of 1% in basal area removal, we generated all possible combinations with the other variables.

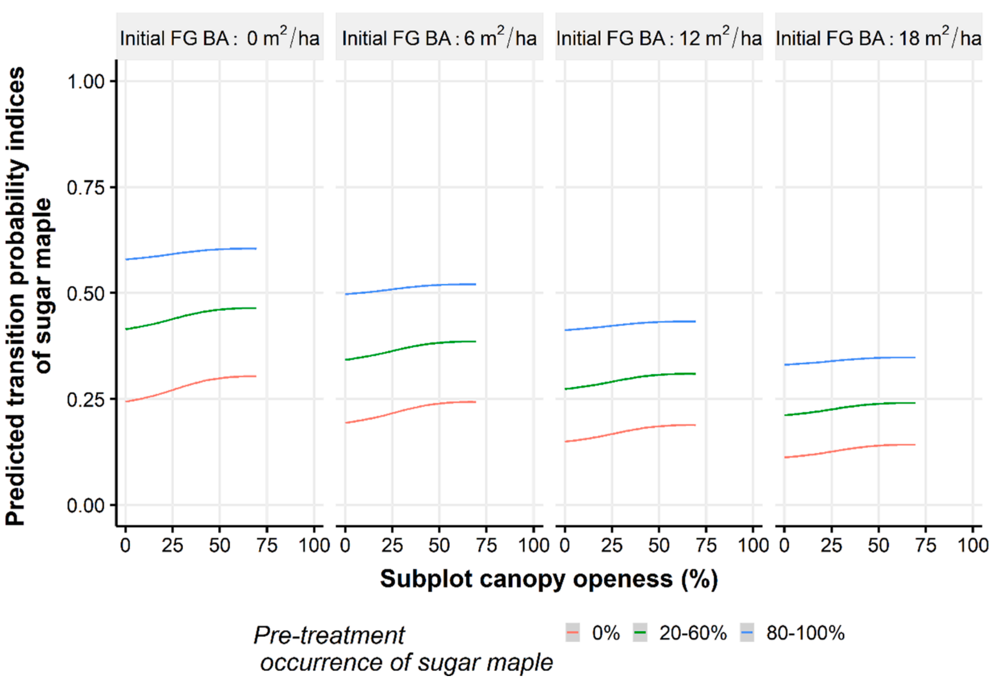

Fig. 3.6. Predictions of transition probability indices for sugar maple as a function of subplot canopy openness per class of sugar maple pre-treatment occurrence in the advance seedling regeneration and initial beech basal area (FG BA). Because we predicted transition probabilities from a simulated database, no confidence intervals are presented. We divided the observed range of subplot canopy openness (0 to 70%) into four sections and calculated the mean basal area removal for each of them. For a simulated subplot canopy openness of 0 to 17.5%, the mean basal area removal used was 35%. For a simulated subplot canopy openness of 17.5 to 35%, 35 to 52.5% and 52.5 to 70%, the mean basal area removal was 45%, 53% and 54%, respectively. For every jump of 1% in subplot canopy openness, we generated all possible combinations with the other variables.

effect on the likelihood of transition toward sugar maple dominance in the regeneration layer. Specifically, the transition probability of sugar maple was predicted to increase by about 0.1 with an increase in basal area removal from 25 to 80% (Fig. 3.5). This is not negligible, considering that the median value is 0.12 for sugar maple and 0.25 for beech (Fig. 3.4). Since species-specific transition probability indices are relative to each other in the same subplot (the sum of all species-specific transition probability indices equals one), an increase in the transition probability index for sugar maple necessarily implies a decrease in other species, and vice and versa. We can therefore speculate that beech, the most abundant species, is most disadvantaged by increases in transition probability indices for sugar maple. Because sugar maple transition probability increased with increasing harvest intensity without reaching a plateau, more severe disturbances or systems that create larger spatially heterogeneity than the single-tree selection system appear to favour sugar maple dominance over beech (Beaudet et al., 2011, 1999; Poulson and Platt, 1996).

Two factors appear to influence the response of sugar maple to an increase in basal area removal (i.e., the slope of the predictions between a basal areal removal of 0% and 80% in Fig. 3.5): the occurrence of sugar maple in the advance seedling regeneration and initial beech basal area. First, opening the canopy where the occurrence of sugar maple in the advance seedling regeneration is high (i.e. occurrence >= 80%) does not have a great impact on sugar maple dominance. In such situations, sugar maple transition probability indices tended to remain high (>0.5) regardless of the harvest proportion, especially when the initial beech basal area was low (Fig. 3.5). This supports the finding of Forrester et al. (2014) that when sugar maple advance regeneration is predominant before canopy opening, it is likely to remain so after harvest. Secondly, When initial basal area of beech is high, the positive effect of increased harvest intensity is diminished because the necessarily greater removal of beech stimulates fast-growing root suckering (Forrester et al., 2014; Jones and Raynal, 1988) (Fig. 3.5). When the initial beech basal area is high, increasing light availability is clearly not sufficient to reverse the dominance of the species.

Our results showed a more limited effect of subplot canopy openness than of basal area removal, especially when the initial beech basal area was high. This might be explained by both the timing and the resolution of the measurements. Basal area removal was measured in the centre of the plot while canopy openness was measured more locally with assessments made in each subplot. Because of the spatial variation in canopy removal after a partial cut (Forget et al., 2007), increasing basal area removal will also tend to increase the proportion of subplots in which there is a significant increase in canopy openness. At this more local scale, it is possible that the beneficial effect of the opening for the regeneration success of sugar maple tends to reach its maximum at approximately 50% of canopy openness. This might be related to the silvics of sugar maple, which despite suffering from beech-induced competition remains a shade-tolerant species. Whereas limited light conditions in the understory are expected to favour beech regeneration over sugar maple, there may be no benefit in providing additional light beyond a certain point at which shade intolerant species may thrive. However, it is likely that the post-cut threshold would in fact be higher than the 50% identified in this study because of the timing of the measurement. While basal area removal was assessed immediately after treatment, canopy openness measurements were made only three to six years after treatment. Light availability, especially for seedlings, is known to decrease rapidly after selection cutting in northern hardwood forests (Beaudet et al., 2004). Further research would therefore be required to identify the level of local canopy openness at which the benefits to sugar maple regeneration tend to saturate.

One potential drawback of using high intensity partial cuts is that it may favour the proliferation of competition in subplots subjected to almost fully open conditions. Non-tree vegetation competition increases with canopy openness which can, in turn, limit light and other resource availability for regeneration development (Houston, 2001; Matonis et al., 2011). Surprisingly, this factor was not included in the most parsimonious model for seedling height growth. The fact that we estimated competition cover in classes (0, >0–25%, 25–75%, 75–100%) instead of a more precise assessment as a continuous variable might have affected its relevance in the analysis. Also, it is likely that the radial growth of seedlings was more affected by high levels of competition than height growth.

4.2. Management implications

Our results showed that two pre-treatment measurements are needed to help inform where canopy openings created by silvicultural systems could help favour sugar maple regeneration: 1) the occurrence of sugar maple in the advance seedling regeneration and 2) the pre-treatment basal area of beech. Based on our predicted probability indices model, we recommend using an initial beech basal area threshold of $\sim$ 6 m^2 ha^{-1} to guide silvicultural prescriptions. Below this threshold we observed the strongest effect of basal area removal on transition probability index for sugar maple, particularly when the occurrence of sugar maple in the advance seedling regeneration subplots was below 60%. Where sugar maple occurrence in the advance seedling regeneration is high (>80%) and beech basal area is low (lower than $\sim$ 6 m^2 ha^{-1}), our results suggest it is unnecessary to increase harvest intensity. In such situations, the dominance of sugar maple regeneration and a good transition from seedling to sapling should be observed in such conditions, regardless of basal area removal. Close-to-nature partial cuts with low harvest intensity (Nolet et al., 2014), such as single-tree selection cuts (Lorimer, 1989; Tubbs, 1977), are recommended in this case.

In cases where the initial basal area of beech is higher than $\sim$ 6 m^2 ha^{-1}, our results suggest that increasing the harvest intensity in partial cuts will likely not suffice to favour sugar maple dominance. As an alternate solution, Nolet et al. (2015), Nyland et al. (2006) and Bose et al. (2018) recommended the application of a combination of harvesting operations to suppress beech advance regeneration. In such cases, the removal of unmerchantable beech poles and saplings could be integrated into harvesting prescriptions (Nyland et al., 2006).While recognizing the importance of protecting beech trees resistant to the beech bark disease for the species' future (Houston, 2001), our results confirm the need to maintain low levels of beech basal area in northern hardwood forests. Otherwise, the feedback mechanism, whereby high beech basal area leads to higher dominance in regeneration, is likely to be maintained (Giencke et al., 2014). Over the next cutting cycle, the combined effects of a high basal area removal and advance seedling regeneration suppression might then be enough to tip the regeneration dynamics in favour of sugar maple.

Current knowledge on this issue clearly shows that there is no single, wall-to-wall solution that can be efficiently applied but rather a diversity of targeted measures that can be adapted to local conditions. Efficiently addressing the increase of beech abundance in regeneration will require refining and enhancing commercial planning and harvesting in northern hardwoods (Messier et al., 2019; Puettmann et al., 2009). Understory composition must be considered when planning and operating commercial partial cuts and managing for this understory is as important. In general, using only basal area removal to manage light availability for the understory is not sufficient to favour sugar maple.

Moreau et al. (2020) found that 42% of the overstory trees remaining after single tree selection cuts did not experience any change in their competitive environment within a 6-meter radius. In our study, large untreated areas were observed, and several subplots had close-to-zero canopy openness. Specific harvesting methods can be used to avoid leaving areas unaltered after partial cut, provided this is not the result of a deliberate choice. Tree marking can on the one hand be planned to create more heterogeneity in gap size, and create targeted, larger openings specifically where sugar maple seedlings undergo strong competition with beech. The use of secondary trails, directional felling and skid trail layouts can on the other hand provide efficient protection measures for advance regeneration in parts of forest stands where sugar maple seedlings dominate the understory (Bose et al., 2020; Hartmann et al., 2009; Moreau et al., 2020; Naghdi et al., 2016). The development of tools for multitreatment planning, such as that proposed by Lussier and Meek (2014) and Labelle et al. (2018), which consists of applying fine-scale adjustments to silvicultural treatments according to the conditions prevailing locally within a single stand, could help operationalize these recommendations. .

4.3. Limitations of the approach

The recommendations made in this study are based on empirical evidence. Caution must be used in applying them in the context of the increasing effects of global changes on forests. Such changing conditions may alter the outcome of silvicultural measures applied to the same forest type in the future. In our analysis, the calendar year of the first growing season after treatment was retained as a variable influencing sugar maple transition probabilities. This implies that results may vary according to changes in yearly conditions that currently remain unquantified. It is also possible that the forest tent caterpillar outbreaks that affected part of our sampling area from 2016 to 2018 could have affected our results. However, Fortin and Maufette (2002) found that leaves from the upper crown were mostly affected by such outbreaks. Although the phenomenon may have increased light availability in the understory at the beginning of the growing season, we believe this only had a marginal effect on light conditions during most of the growing seasons covered by our study.

Further research is required to elucidate the effects of such changes in yearly conditions so that the response of the forest to future partial cuts can be better anticipated. Moreover, increasing sugar maple seedling transition to sapling does not necessarily imply an increase in sugar maple sapling to pole transition. Hence, a meticulous monitoring followed by beech sapling suppression, if required, should be considered to ensure a successful transition from saplings to poles for sugar maple. Also, other issues than beech competition must be considered when making silvicultural prescriptions that promote the future health, quality and resilience of northern hardwood forests. We should not manage forests only for sugar maple, but for a greater species and structural diversity to ensure that forests will maintain their resistance and resilience in a rapidly changing socio-environmental context (Messier et al., 2019).

5. Conclusions

While total sugar maple regeneration was more abundant than beech across our study, the greater abundance of beech among tall seedling and sapling classes suggests a greater likelihood of increasing dominance in the future. Our results suggest increasing harvest intensity may favour sugar maple over beech when sugar maple constitutes < 60% of advance seedling regeneration and the overstory contains low beech abundance (basal area < $6m^2ha^{-1}$). In all other conditions, the within-stand variation of the harvest intensity within the spectrum of partial harvesting, that is, from 0% to 80% of basal area removal, had very little effect on the transition probability indices for sugar maple. Considering that beech leaf disease might move in from southern Ontario (Ewing et al., 2019), the suggested threshold of 6 m^2 ha^{-1} of beech initial basal area must be used with care. Beyond determining a threshold, our results showed that more intensive silvicultural tools, such as the mechanical removal of beech seedlings and saplings, are necessary to limit the regeneration success of beech. Considering that 1) global changes are affecting northern hardwoods, 2) the species composition of the forest regeneration is not static over time (Knapp et al., 2019), and 3) we did not assess the transition from the sapling to pole stage in this study, we conclude that long-term monitoring of the effects of commercial silvicultural systems is still necessary to gain more knowledge of the issue. A better understanding of forest understory dynamics will provide forest managers the tools to manage the forest of tomorrow and thus enhance its resilience to face uncertain changes (Webster et al., 2018). Based on our results, despite their high abundance, sugar maple seedlings could be overtaken by beech saplings in the absence of interventions. A flexible and refined diagnostic process and silvicultural prescription are therefore necessary to ensure the future of sugar maple, and more generally the resilience of northern hardwood forests.

CRediT authorship contribution statement

Émilie St-Jean: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Project administration, Funding acquisition. **Sébastien Meunier:** Conceptualization, Methodology, Writing – review & editing. **Phillipe Nolet:** Methodology, Formal analysis, Writing – review & editing. **Christian Messier:** Methodology, Writing – review & editing. **Alexis Achim:** Conceptualization, Methodology, Validation, Formal analysis, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foreco.2021.119607.

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